

Engineering Mathematics by Prof. Nitin

Practice Exercise for Sem – IV

Module 1 - Linear Algebra Matrix Theory-I

- 1) Find eigenvalues of a) $A^3 + 5A + 8I$ b) A^{-1} c) $\text{adj}(A)$ if matrix $A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$.
- 2) Find eigenvalues of $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$, $\begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$, $\begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 3 & 3 & 4 \end{bmatrix}$.
- 3) Find the eigenvalues of $3A^3 + 5A^2 - 6A + 2I$ if $A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & 3 & 2 \\ 0 & 0 & -2 \end{bmatrix}$.
- 4) If the matrix $A = \begin{bmatrix} 2 & 1 & -2 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, find eigenvalues of $A^3 + 2A^2 + 3I$.
- 5) If the matrix $A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$, find eigen values of A^t , A^{-1} , $\text{adj}(A)$.
- 6) Find eigenvalues and eigenvectors of the following matrices
- a) $\begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$ b) $\begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$ c) $\begin{bmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$ d) $\begin{bmatrix} -9 & 4 & 4 \\ -8 & 3 & 4 \\ -16 & 8 & 7 \end{bmatrix}$ e) $\begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 3 & 3 & 4 \end{bmatrix}$
- 7) Verify Cayley-Hamilton theorem for the following matrix and find the matrix represented by $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$ if $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$.
- 8) Verify Cayley-Hamilton theorem for the following matrix and find the matrix represented by $A^6 - 6A^5 + 9A^4 - 2A^3 + 12A^2 + 23A - 9I$ if $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$.
- 9) Using Cayley Hamilton Theorem Compute $A^6 - 6A^5 + 9A^4 + 4A^3 - 12A^2 + 2A - I$, where $A = \begin{bmatrix} 3 & 10 & 5 \\ -2 & -3 & -4 \\ 3 & 5 & 7 \end{bmatrix}$.
- 10) Using Cayley Hamilton Theorem Compute $A^7 - 4A^6 - 20A^5 - 35A^4 - 4A^3 - 20A^2 + 33A + I$, where $A = \begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix}$.
- 11) Verify Cayley-Hamilton theorem for the following matrix and find A^4 , A^{-1} if $\begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$, $\begin{bmatrix} 3 & 1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$,
- 12) Show that the matrix A is diagonalizable and find the corresponding transforming (modal) matrix and the diagonal matrix.

$$\begin{bmatrix} 8 & -8 & -2 \\ 4 & -3 & -2 \\ 3 & -4 & 1 \end{bmatrix} = \begin{bmatrix} -9 & 4 & 4 \\ -8 & 3 & 4 \\ -16 & 8 & 7 \end{bmatrix}, \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}, \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}, \begin{bmatrix} -2 & -5 & -1 \\ 3 & 6 & 1 \\ -2 & -3 & 1 \end{bmatrix}$$

Module 2 - Linear Algebra Matrix Theory-II

1) Find the Nullity of the matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 1 & -1 & 1 \\ 2 & 4 & 0 & 6 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 2 & 3 \\ 2 & 2 & 4 & 6 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 3 \\ 2 & 6 \\ -1 & -3 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 1 & 2 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 3 & 4 \end{bmatrix}$$

2) Find the LU decomposition of the matrix

$$\begin{bmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{bmatrix}, \begin{bmatrix} 2 & 4 & 3 & 5 \\ -4 & -7 & -5 & -8 \\ 6 & 8 & 2 & 9 \\ 4 & 9 & -2 & 14 \end{bmatrix}, \begin{bmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{bmatrix}, \begin{bmatrix} 2 & 3 & 1 \\ 6 & 13 & 5 \\ 2 & 19 & 10 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 3 \\ 4 & 8 & 10 \\ 7 & 14 & 22 \end{bmatrix},$$

$$\begin{bmatrix} 3 & 2 & -1 \\ 6 & 5 & 0 \\ -3 & 2 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 1 \\ 4 & 3 & -1 \\ 3 & 5 & 3 \end{bmatrix}.$$

3) Find the Cholesky decomposition of matrix

$$\begin{bmatrix} 25 & 15 & -5 \\ 15 & 18 & 0 \\ -5 & 0 & 11 \end{bmatrix}, \begin{bmatrix} 4 & 12 & -16 \\ 12 & 37 & -43 \\ -16 & -43 & 98 \end{bmatrix}, \begin{bmatrix} 9 & 6 & 3 \\ 6 & 8 & 6 \\ 3 & 6 & 14 \end{bmatrix}, \begin{bmatrix} 4 & 10 & 8 \\ 10 & 26 & 26 \\ 8 & 26 & 61 \end{bmatrix}, \begin{bmatrix} 25 & 15 & -5 \\ 15 & 18 & 0 \\ -5 & 0 & 11 \end{bmatrix}$$

4) Find Singular Value decomposition of the matrix

$$\begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix}, \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 1 \\ \sqrt{2} & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}$$

Module 3 – Statistics for Data Analysis

Q1. Let's say we have the following data representing the marks of students in a class, grouped into intervals. Calculate the mean and the median.

Marks interval	0-10	10-20	20-30	30-40
Frequency f_i	3	5	8	4

Q2. The following data represents the marks of students in a class, grouped into intervals. Calculate the mean and the median.

Marks	40-50	50-60	60-70	70-80	80-90
No. of students	28	29	33	31	26

Q3. The following table presents a frequency distribution of a dataset. Calculate the first quartile (Q_1), the second quartile (Q_2), the third quartile (Q_3), the mean, and the variance from the data below:

Class Interval (CI)	Frequency (F)
0-50	5

50-100	10
100-150	3
150-200	4
200-250	0
250-300	15
300-350	6
350-400	5
400-450	2

Q4.. A factory measures the time taken (in minutes) by workers to complete a task. The data is given below. Determine the quartiles Q1, Q2, and Q3 and IQR.

Time	5-15	15-25	25-35	35-45	45-55	55-65
Frequency	4	6	10	14	12	8

Q5. Find Q1, Q2, Q3 and interquartile range and range for the following data:

Salary(in thousands)	20-25	25-30	30-35	35-40	40-45	45-50	50-55	55-60
No. of workers	5	10	15	25	65	40	23	10

Q6. Find Q1, Q2, Q3 and interquartile range and range for the following data:

Class Interval	0-10	10-20	20-30	30-40	40-50
Frequency f_i	5	8	12	20	10

Q7. The following table represents the daily sales (in rupees) of a small shop over a month. Find interquartile range of this data.

Sales (in Rs.)	100-200	200-300	300-400	400-500	500-600	600-700
Frequency	3	7	12	18	10	5

Q8. The marks obtained by students in a test are given below. Find variance and standard deviation of this data.

Marks	0-10	10-20	20-30	30-40	40-50	50-60
Frequency	4	8	10	15	8	5

Q9. The table below shows the number of hours studied and the corresponding exam scores of 6 students. Find covariance of the data.

Study hours	2	4	6	8	10	12
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Exam score	50	55	65	70	80	90
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Q10. The following table represents the heights (in cm) of students in a school:

Height Interval (cm)	Frequency (f)
140 - 150	6
150 - 160	10
160 - 170	15
170 - 180	12
180 - 190	7

Find the variance and the standard deviation of the data.

Q11. Find Kendall's rank correlation coefficient for the given data.

X ranks	1	2	3	4	5	6
Y Ranks	3	1	4	2	6	5

Q12. The ranks of 6 students in Math and Science are given below. Find Kendall rank correlation coefficient of the following data.

Ranks in Maths	1	2	3	4	5	6
Ranks in English	2	4	1	6	3	5

Q13. Find Kendall's rank correlation coefficient for the given data.

X	33	61	20	19	40
Y	26	36	65	25	35

Q14. Calculate the Kendall's Tau rank correlation coefficient for the following data:

X	5	11	2	10	8	7
Y	4	10	13	5	8	9

Module 4 Probability Distribution

Q1. There are 5% defective items in a large bulk of items. What is the probability that a sample of 10 items will include not more than one defective item?

Q2. In the last 100 years, there have been 93 earthquakes measuring 6.0 or more on the Richter scale. What is the probability of having 3 earthquakes in the same year that all measure 6.0 or more?

Q3. In a certain factory producing cycle tyres, there is a small chance of 1 in 500 tyres to be defective. The tyres are supplied in lots of 10. Using Poisson distribution calculate the approximate number of lots containing no defective, one defective and two defective tyres, respectively in a consignment of 10,000 lots.

Q4. A transmission channel has a one digit error probability 0.01 . Calculate the probability of more than 1 error in 10 received digits using (1) Binomial Distribution (2) Poisson Distribution

Q5. The incomes of a group of 10,000 people were found to be normally distributed with a mean of Rs. 520 and a standard deviation of Rs. 60. Find i) the number of people having income between Rs. 400 and Rs. 550, and ii) the lowest income of the richest 5% of the people.

[Area under standard normal curve from $z = 0$ to 2 is 0.6915, from $z = 0$ to 0.5 is 0.0228, area from 0 to 0.51 is 0.1950 , area from 0 to 0.45 is 0.1736 , area for 0 to 0.56 is 0.2123 , area from 0 to 0.55 is 0.2088]

Q6. Most graduate schools of business require applicants for admission to take the Graduate Management Admission Council's GMAT examination. Scores on the GMAT are roughly normally distributed with a mean of 527 and a standard deviation of 112. What is the probability of an individual scoring above 500 on the GMAT?

Q7. The lifetime of a light bulb is X hours, where X can be modelled by an exponential distribution with parameter $\lambda = 0.0125$.

a) Find the mean and variance of the lifetime of a light bulb.

b) Find the probability that the lifetime of a bulb is:

(i) less than 100 hours; (ii) between 50 hours and 150 hours.

Q8. The lifetime (in hundreds of hours) of a certain type of vacuum tube has a Weibull distribution with parameters α and β . Compute the following:

a. $E(X)$ and $V(X)$

b. $P(X \leq 5)$

c. $P(1.8 \leq X \leq 5)$

Q9. In a sample of 1000 cases. The mean of a certain test is 14 and SD is 2.5. Assuming the distribution to be normal, find

1) How many students score between 12 and 15?

2) How many score above 18?

Note: Area under the z curve from $z = 0$ to $z = 0.8$ is 0.2881, from $z = 0$ to $z = 0.4$ is 0.1554, from $z = 0$ to $z = 1.6$ is 0.4452, from $z = 0$ to $z = 0.51$ is 0.1950, area from $z = 0$ to $z = 0.45$ is 0.1736 , area for $z = 0$ to $z = 0.56$ is 0.2123.

Q10. A magnetic disk is exposed to corrosive gas. Life time of a disk follows Weibull distribution with parameters $\lambda = 300$ and $k = 0.5$. What will be the probability of it
i) failing before 500 hours? ii) lasts 600 hours or more?

Q11. In a distribution exactly normal 7% of items are under 35 and 89% are under 63. What are mean and standard deviation? And also find the probability that an item selected at random lies between 45 and 56.

[Given that area from 0 to 0.51 is 0.1950, area from 0 to 0.45 is 0.1736, area for 0 to 0.56 is 0.2123, area from 0 to 0.55 is 0.2088 and for the area 0.43 the $z = 1.48$, for the area 0.07 the $z = 0.18$, for the area 0.39 the $z = 1.23$]

Q12. The life time of a rechargeable battery under constant usage condition, follows a Weibull distribution with parameters shape parameter = 2 and Scale parameter $R = 10$.

- a) What is the probability that a battery is still functional at time $X = 20$?
- b) What is the mean time to failure?
- c) What is the probability that a battery is still functional at time $X = 20$ given that the battery is functional at time $X = 10$?

Q13. It is known that the probability of an item produced by a certain machine being defective is 0.05. If the produced items are sent to the market in packets of 20, find the number of packets containing exactly, at least, and at most 2 defective items in a consignment of 1000 packets.

Q14. The time (in hours) required to repair a watch is exponentially distributed with parameter $\lambda = 1/2$.

- (I) What is the probability that the repair time exceeds 2 hours?
- (II) Given that its duration exceeds 8 hours, what is the probability that it will take more than 11 hours in total?

Module 5 Sampling Theory

Q1. A random sample of 50 items gives the mean 6.2 and variance 10.24. Can it be regarded as drawn from a normal population with mean 5.4 at 5% level of significance?

(Table value t distribution for DOF 49 and LOS 5% is 1.96)

Q2. The following are the gain in weights of cows fed on two types of diets X and Y.

Diet X: 30 37 35 37 29 19 37

Diet Y: 29 39 27 35 47 37 45 35 37 40

Test at 5% level of significance whether the two diets differ as regards their effect on mean increase in weight. (Table value t distribution for DOF 15 and LOS 5% is 2.131)

Q3. Investigate the association between the darkness of eye colour in father and son from the following data.

Colour of Son's eye	Colour of father's eyes			
		Dark	Not Dark	Total
	Dark	48	90	138
	Not Dark	80	782	862
	Total	128	872	1000

(Table value for χ^2 value at DOF 1 and LOS 5% is 3.84)

Q4. 300 digits were chosen at random from a table of random numbers. The frequency of digits was as follows. Using χ^2 test examine the hypothesis that the digits were distributed in equal numbers in the table (take LOS 5%)

(Note: Critical value at 5% LOS with at dof 1 is 3.841, at dof 3 is 7.815, at dof 9 is 16.92)

Digit	0	1	2	3	4	5	6	7	8	9
Frequency	28	29	33	31	26	35	32	30	31	25

Q5. A total number of 3759 individuals were interviewed in a public opinion survey on a political proposal. Of them, 1872 were men and the rest were women. A total of 2257 individuals were in favor of the proposal, and 917 were opposed to it. A total of 243 men favour were undecided, and 442 women opposed the proposal. Do you justify or contradict the hypothesis that there is no association between gender and Opinion?

[The table value of χ^2 at LOS 0.05, and at dof 2 is 5.991, at dof 1 is 3.841, at dof 3 is 7.815]

Q6. Following results were obtained when body fat % of men and women who work out in gym is measured. Is the difference between

the means of the samples significant? (take 5% LOS) (Note: Critical (take 5% LOS) (Note: Critical value at 5% LOS with DOF 21 is 2.080, with DOF 22 is 2.074, with DOF 23 is 2.069)

	Sample Size	Mean	S.D.
Men (x_1):	13	14.95	6.84
Women (x_2):	10	22.29	5.32

Q7. Two batches of 12 animals each are given test of inoculation. One batch was inoculated and the other was not. The number of dead and surviving animals is given below for both the cases. Can the inoculation be regarded as effective against the disease?

	Dead	Surviving
Inoculated	2	10
Non inoculated	8	4

[The table value of χ^2 at 5% LoS at dof 1 is 3.841, at DoF 2 is 5.991, at DoF 3 is 7.815 respectively.]

Q8. The means of two random samples of size 9 and 7 are 196.42 and 198.82 respectively. The sum of the squares of deviations from the means is 26.94 and 18.73 respectively. Can the samples be considered to have been drawn from the same population?

[The table value of t at 5% Level of significance for degree of freedom 16,15,14,13 is 2.120 , 2.131 , 2.145 , 2.650 respectively.]

Q9. A commuter in a large city can travel to work car, bicycle or bus. Below are the results i(n min) by each method

Car	45	27	33	31
Bicycle	38	34	43	42
Bus	41	26	35	46

Carry out analysis of variance whether there are differences in the mean journey times for the three methods of transport. (Table value of F-distribution at DOF $F(3,16), F(2,9), F(3,11)$ at LOS 0.05 are 3.2389, 4.26, 3.59 respectively).

Q10. Test the differences between the means of following groups using one-way ANOVA test. (Table value of F-distribution at DOF (3,16) and LOS 0.05 is 3.2389).

A	B	C	D
8	12	18	13
10	11	12	9
12	9	16	12
8	14	6	16
7	4	8	15

Q11. To assess the significance of possible variations in performance in a certain test between the convent schools of a city, a common test was given. The results are given below. Make an analysis of variance of the data at 5% of LOS. [Table value of F distribution at DOF $F(3,16), F(2,12), F(2,9), F(3,11)$ at LOS 0.05 are 3.2389, 3.89, 4.26, 3.59 respectively]

A	B	C	D
8	12	18	13
10	11	12	9
12	9	16	12
8	14	6	16
7	4	8	15

Q12. A car manufacturer wants to compare the fuel efficiency (miles per gallon) of three car brands (Brand A, Brand B, and Brand C). The fuel efficiency data for 5 cars of each brand is as follows:

Brand A	Brand B	Brand C
9	13	14
11	12	13
13	10	17
9	15	7
8	5	9

Perform a One-Way ANOVA to test if there is a significant difference in the mean fuel efficiency among the three brands at a 5% significance level.

[Table value of F distribution at DOF $F(3,16), F(2,12), F(2,9), F(3,11)$ at LOS 0.05 are 3.2389, 3.89, 4.26, 3.59 respectively].

Module 6 – LPP and NLPP

1) Convert the following LPP in canonical and standard form

a) Maximize $Z = 4x_1 + 3x_2$
Subject to $2x_1 + x_2 \leq 8$
 $x_1 + 3x_2 \geq 6$
 $x_1, x_2 \geq 0$

b) Minimize $Z = 3x_1 + 5x_2$
Subject to $x_1 + 2x_2 \geq 4$
 $3x_1 + x_2 \leq 6$
 $x_1, x_2 \geq 0$

c) Minimize : $Z = 2x_1 + 4x_2$
Subject to $x_1 + 2x_2 \geq 3$
 $x_1 + x_2 \leq 5$
 $x_1, x_2 \geq 0$

d) Maximize $Z = 4x_1 - x_2$
Subject to $2x_1 - x_2 \leq 2$
 $x_1 + 2x_2 \geq 5$
 $x_2 \geq 0$

2) Find all the basic-feasible solution of the following LPP

a) Maximize $Z = 3x_1 + 2x_2$
Subject to $x_1 + x_2 \leq 6$
 $2x_1 + x_2 \leq 10$
 $x_1, x_2 \geq 0$

b) Maximize $Z = 2x_1 + 3x_2 + 4x_3 + 7x_4$
Subject to $2x_1 + 3x_2 + x_3 + 4x_4 = 8$
 $x_1 - 2x_2 + 6x_3 - 7x_4 = -3$
 $x_1, x_2, x_3, x_4 \geq 0$

c) Minimize $Z = 5x_1 + 4x_2$
Subject to $x_1 + 2x_2 \leq 8$
 $2x_1 + x_2 \leq 10$
 $x_1, x_2 \geq 0$

d) Maximize $Z = x_1 + x_2 + 3x_3$
Subject to $x_1 + 2x_2 + 3x_3 = 9$
 $3x_1 + 2x_2 + 2x_3 = 5$
 $x_1, x_2, x_3 \geq 0$

e) Maximize $Z = 2x_1 - 2x_2 + 4x_3 - 5x_4$
Subject to $x_1 + 4x_2 - 2x_3 + 8x_4 \leq 2$
 $-x_1 + 2x_2 + 3x_3 + 4x_4 \leq 1$
 $x_1, x_2, x_3, x_4 \geq 0$

f) Maximize $Z = 3x_1 + 2x_2$
Subject to $x_1 + x_2 \leq 6$
 $2x_1 + x_2 \leq 10$

- $x_1, x_2 \geq 0$
- 3) Solve the following LPP Using Simplex method
- a) Maximize $Z = 3x_1 + 5x_2$
 Subject to $x_1 + 2x_2 \leq 4$
 $3x_1 + 2x_2 \leq 6$
 $x_1, x_2 \geq 0$
- b) Maximize $Z = 4x_1 + 3x_2 + 6x_3$
 Subject to $2x_1 + x_2 + x_3 \leq 8$
 $x_1 + 3x_2 + 2x_3 \leq 12$
 $x_1 + 2x_2 + 4x_3 \leq 16$
 $x_1, x_2, x_3 \geq 0$
- c) Maximize $Z = x_1 - 3x_2 + x_3$
 Subject to $3x_1 - x_2 + 2x_3 \leq 7$
 $2x_1 + 4x_2 \geq -12$
 $-4x_1 + 3x_2 + 8x_3 \leq 10$
 $x_1, x_2, x_3 \geq 0$
- d) Maximize : $Z = 5x_1 + 4x_2$
 Subject to $x_1 + x_2 \leq 6$
 $2x_1 + x_2 \leq 8$
 $x_1, x_2 \geq 0$
- e) Maximize $Z = 4x_1 - x_2$
 Subject to $x_1 + 2x_2 \leq 4$
 $2x_1 + 3x_2 \leq 12$
 $x_1 - x_2 \leq 3$
 $x_1, x_2 \geq 0$
- f) Maximize : $Z = 3x_1 + 5x_2 + 4x_3$
 Subject to $2x_1 + 3x_2 \leq 8$
 $2x_1 + 5x_2 \leq 10$
 $3x_1 + 2x_2 + 4x_3 \leq 15$
 $x_1, x_2, x_3 \geq 0$
- g) Maximize $Z = x_1 + 5x_2 + x_3$
 Subject to $x_1 + 3x_2 + x_3 \leq 4$
 $2x_2 + x_3 \leq 2$
 $x_1, x_2, x_3 \geq 0$
- h) Maximize $Z = 3x_1 + 2x_2 + 5x_3$
 Subject to $x_1 + 2x_2 + x_3 \leq 430$;
 $3x_1 + 2x_3 \leq 460$;
 $x_1 + 4x_2 \leq 420$.
 $x_1, x_2, x_3 \geq 0$.
- i) Maximize $Z = x_1 + 5x_2 + x_3$
 Subject to $x_1 + 3x_2 + x_3 \leq 4$
 $2x_2 + x_3 \leq 2$
 $x_1, x_2, x_3 \geq 0$
- 4) Find the relative maximum or minimum of the function
- a) $z = x_1^2 + x_2^2 - 4x_1 - 8x_2 + 100$.
- b) $z = x_1 + 2x_3 + x_2x_3 - x_1^2 - x_2^2 - x_3^2 + 50$.
- c) $Z = x_1^2 + x_2^2 + x_3^2 - 6x_1 - 8x_2 - 10x_3 + 100$
- d) $Z = x_1^2 + x_2^2 - 4x_1 - 6x_2 + 13$

5) Solve following NLPP using Lagrange's method

- a) Optimize $z = 6x_1^2 + 5x_2^2$
Subject to $x_1 + 5x_2 = 7$
 $x_1, x_2 \geq 0$
- b) Optimize $Z = x_1^2 + x_2^2$
Subject to $x_1 + x_2 = 1$
 $x_1, x_2, x_3 \geq 0$
- c) Optimize $Z = 4x_1 + 8x_2 - x_1^2 - x_2^2$
Subject to $x_1 + x_2 = 4$
 $x_1, x_2, x_3 \geq 0$
- d) Optimize $Z = 4x_1 + 8x_2 - x_1^2 - x_2^2$
Subject to $x_1 + x_2 = 2$
 $x_1, x_2 \geq 0$
- e) Optimize $Z = 2x_1 + 6x_2 - x_1^2 - x_2^2 + 14$
Subject to $x_1 + x_2 = 4$
 $x_1, x_2 \geq 0$
- f) Optimize $Z = x_1x_2$
Subject to $x_1^2 + x_2^2 = 10$
 $x_1, x_2 \geq 0$